Chapter 6 : Introduction to inference

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⇒ The purpose of statistical inference is to draw conclusions from data, conclusions take into account the natural variability in the data.

⇒ Formal inference relies on probability to describe chance variation.

↳ We can then correct our judgement by calculation.

EXAMPLE 6.1 Compare TV Commercials

Suppose we show a new TV commercial and the present commercial to 20 consumers each. Twelve of those who see the new ad declare an interest in buying the product, versus only 8 who watch the current version. Is the new commercial more effective?

⇒ In the above example, probability calculation tells us a difference this large or larger between the results in two groups would occur about one time in five simply because of chance variation.

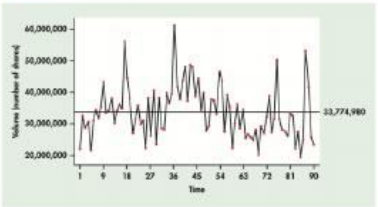
↳ Since this probability is not small, it's reasonable to conclude that the difference is due to chance variations, rather than a real difference.

EXAMPLE 6.2 Pattern in Trading Volumes

Investors and stock analysts are continually seeking ways to understand price movements in stocks. One factor many have considered in the prediction of price movements is trading volume, that is, the quantity of shares that change owners. In studying the relationship between trading volumes and price movements, an understanding of the trading-volume process becomes a source of interest. Are trading volumes random from trading period to trading period? Or is there some pattern in the trading volumes over

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FIGURE 6.1 Time series plot of daily trading volumes of AT&T stock, with the mean trading volume indicated.



time? Figure 6.1 is a time plot of daily trading volumes for AT&T stock from January 2 to May 12, 2009, with the mean trading volume indicated. Do the volumes appear to be randomly distributed around the mean trading volume over time? More specifically, could you simulate the trading-volume process with the process of flipping a fair coin, with heads being an observation below the mean and tails being an observation above the mean? Or does it appear that the volumes move in patterns around the mean level? In particular, are there "strings" of observations either all below or all above the mean? As you will learn in Chapter 13, one approach to the analysis of these data, known as runs test analysis, indicates that observing a sequencing of observations as patterned as or more patterned than that shown in Figure 6.1 would occur only 2.1% of the time if, in fact, trading volumes are truly random over time.

↳ Because this chance is fairly small, we are inclined to believe that the trading volumes are not random over time.

⇒ We'll see two most prominent types of formal statistical inference

1) The confidence intervals used for estimating the value of population parameter.
2) tests of significance, which assesses the evidence for a claim.

⇒ Both type of inference is based on the sampling distributions of statistics.

↳ i.e, both report probabilities that state what would happen if we used the inference method many time.

⇒ Formal inference procedures are based on sampling distributions. This means that they require a probability model that describes how the data are generated.

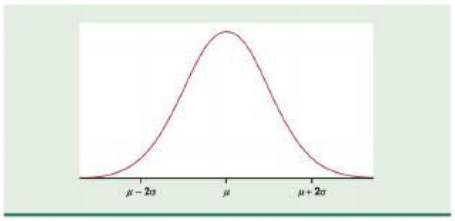
EXAMPLE 6.3 A Probability Model

Consider sampling from a Normal population. The mean is μ and the standard deviation is σ. A picture of this model is given in Figure 6.2.

⇒ We're concerned about general ideas about inference. Will illustrate these ideas in the setting of inference for means.

⇒ The sampling distribution for the mean is the sampling distribution that is relevant for the inference.

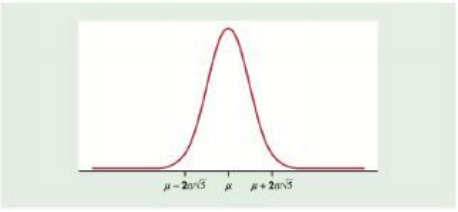
FIGURE 6.2 The probability model for Example 6.3.



EXAMPLE 6.4 The Sampling Distribution

Suppose we take a random sample of size 5 from the Normal population in Example 6.3. From Section 4.4 we know that the mean of this sampling distribution is μ and the standard deviation is $\sigma/\sqrt{n}$, or $\sigma/\sqrt{5}$ for our sample size of $n = 5$. A picture of this sampling distribution is given in Figure 6.3.

FIGURE 6.3 The sampling distribution for Example 6.4.



⇒ We often argue that our inference is based on assuming an underlying process that generates the data rather than some random mechanism associated with its collection.

⇒ Because we're concentrating on the reasoning of statistical inference in this chapter, we temporarily make a simplifying assumption that we know the standard deviation σ of population.

↳ In application based on large sample sizes, this assumption is not unrealistic since the greater amount of information provides us with reasonable value of σ.

Estimating with Confidence:

⇒ We use the term population to refer to the entire collection of interest.

EXAMPLE 6.5 Health Insurance

DATA FILE HMO

The Kaiser Family Foundation (Kaiser) and the Health Research & Educational Trust (HRET) conduct an annual national survey of firms on a variety of health coverage issues, including premiums and employee contributions.[1] The 2008 survey included 2832 randomly selected public and private firms with three or more employees. Firms are categorized as "large" if they have 200 or more employees. In the sample, nearly 100% of large firms offer health benefits.

The survey considers different health plans such as health maintenance organizations (HMOs),

preferred-provider organizations (PPOs), point-of-service plans (POS), and high-deductible health plans with a savings option (HDHP/SO). Let us consider the mean monthly cost of the premium for single-coverage HMO plans for workers in large firms. For this situation, we find 453 firms in the sample. For these firms, the mean monthly premium is $\bar{x} = \$405.02$. What can we say about μ, the mean monthly single-coverage premium per worker for all large firms in the United States?

⇒ The sample mean $\bar{x}$ is the natural estimator of the unknown population mean μ.
↳ We also know $\bar{x}$ is an unbiased estimator of μ.
↳ Unbiased means there is no systematic tendency to constantly over/under estimate.

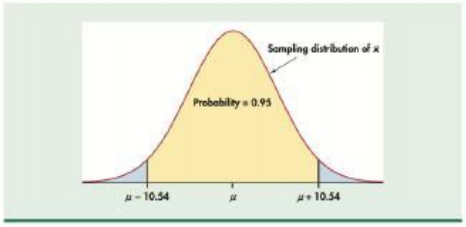
⇒ The law of large numbers says that the sample mean must approach the population mean as the size of sample grows.
↳ The value $\bar{x} = 405.02$ appears to be reasonable estimate of mean monthly premium μ for all large firms.
↳ Q) But how reliable is this estimate?
↳ A second sample would surely not give 405.02 again.

⇒ An estimate without an indication of its variability is of limited value.

- Statistical confidence :-

⇒ By central limit theorem we know that, if the entire population has a mean μ and standard deviation σ, then in repeated samples of size 453 the sample mean $\bar{x}$ approximately follows the $N(\mu, \sigma/\sqrt{453})$ distribution.
↳ Suppose that the true standard deviation σ is equal to the sample standard deviation $s = 112.08$. (Although not true, but will give us accurate results for samples as large as 453).
↳ In the next chapter we'll learn how to proceed when σ is not known.

FIGURE 6.4 In 95% of all samples, $\bar{x}$ lies within ±10.54 of μ. So μ also lies within ±10.54 of $\bar{x}$ in those samples.



⇒ for now were interested in statistical reasoning.

$$\therefore \mu_{\bar{x}} = \frac{\$112.08}{\sqrt{453}} = \$5.27.$$

⇒ Now we can talk about estimating μ. Consider the below line of thought that is reflected in (figure 6.4).

1) The 68-95-99.7 rule states that the probability is about 0.95 that $\bar{x}$ is within 10.54 (two standard deviations of $\bar{x}$) of the population mean monthly premium μ.

2) To say that $\bar{x}$ lies within 10.54 of μ is the same as saying

that μ is within 10.54 of $\bar{x}$. ($a=b \Rightarrow b=a$).

3) So 95% of the samples will capture the true mean μ in the interval from $\bar{x}-10.54$ to $\bar{x}+10.54$.

⇒ We've simply restated a fact about sampling distribution of $\bar{x}$.

⇒ The language of statistical inference uses this fact about what would happen in the long run to express our confidence in any results of any one sample.

⇒ We say that we are 95% confident that the unknown mean monthly premium for all large firms lies between

$\bar{x} - 10.54$ & $\bar{x} + 10.54$

$= [\$394.48, \$415.56]$

⇒ There are only two possibilities with this:

either

1) The interval between \$394.48 and \$415.56 contains the true μ.

or

2) Our SRS was one of the few samples for which $\bar{x}$ is not within 10.54 of the true μ. Only 5% of all samples give such inaccurate results.

⇒ We cannot know whether our sample is one of the 95% for which the interval catches $\bar{x} \pm 10.54$ or one of the unlucky 5%.

↳ We say "We arrived at these numbers by a process that gives correct results 95% of time".

- Confidence intervals :—

⇒ A level C confidence interval for a parameter has two parts:

↳ An interval calculated from the data, usually of the form

estimate ± margin of error.

↳ A confidence level C, which gives the probability that the interval will capture the true parameter value in repeated samples.

- Confidence interval for a population mean:

⇒ We use the sampling distribution of the sample mean $\bar{x}$ to construct a level C confidence interval for the mean μ of a population.

⇒ We assume that the data are an SRS of size n.

⇒ The sampling distribution is exactly $N(\mu, \sigma/\sqrt{n})$ when population has $N(\mu, \sigma)$.

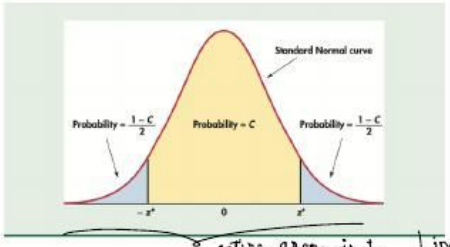
⇒ Our construction of 95% confidence interval began by noting that any normal distribution has probability of about 0.95 of taking a value within $\mu \pm 2\sigma$

⇒ To construct a level C confidence interval, we first catch the central C area under the Normal curve.

⇒ Because all normal distributions are same in standard scale, we can obtain everything we need from the standard Normal curve.

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FIGURE 6.6 The area between the critical values $-z^*$ and z^* under the standard Normal curve is C.



entire area is 1
blue area = $1-C$
individual blue area = $\frac{1-C}{2}$.

⇒ The above figure shows the relationship between the central area C & points z^* that mark off the area.

⇒ Below are the most important values of z^* & corresponding C

z^*	1.645	1.960	2.576
C	90%	95%	99%

⇒ Values z^* that mark off specific areas are called critical values of the standard Normal Distribution. Notice that for C = 95% the table gives $z^* = 1.960$. This is slightly more precise than $z^* = 2$

↳ Remember z-score (z^*) means how many std away a data is from the mean.

- Confidence interval for a population Mean:

⇒ Choose an SRS of size n from a population having an unknown mean μ and a known standard deviation σ. A level C confidence interval for μ is

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$$

(z^*: z-score (how many σ a data is from mean); $\frac{\sigma}{\sqrt{n}}$: new std of the sampling distribution)

⇒ Here z^* is a critical value with area C between $-z^*$ and z^* under the standard Normal curve.

⇒ The quantity $z^* \sigma/\sqrt{n}$ is the margin of error. The interval is exact when the population distribution is Normal, and is approximately correct when n is large.

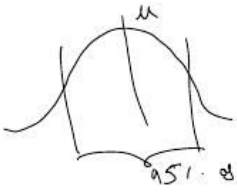
↳ Because sampling distribution is approximately $\sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$,

and only $= N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$ when population has $N(\mu, \sigma)$ distribution.

Q) Why is the interval $\bar{x} \pm 2\frac{\sigma}{\sqrt{n}}$ containing μ with C level of confidence?

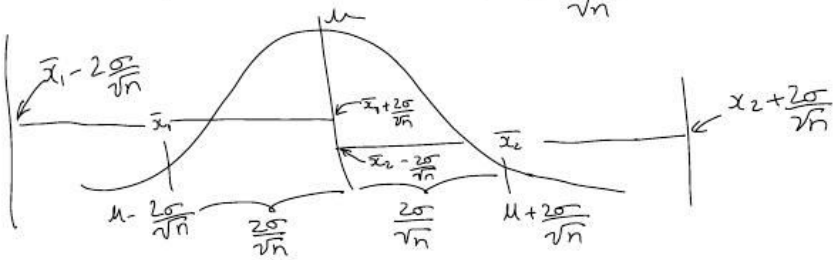
→ The sampling distribution is $\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$, so the mean does happen in sampling distribution, so wide on the right path.

⇒ Now



This checks out for any $\bar{x}$.

⇒ Since multiple $\bar{x}$ makes up the interval, 95% of the $\bar{x}$ will belong to the interval $\mu \pm \frac{2\sigma}{\sqrt{n}}$.



Now the worst case is $\bar{x}$ lies on edges, the interval $\bar{x} \pm \frac{2\sigma}{\sqrt{n}}$ will have μ either right or left of the edge.

↳ Hence even in the worst case the interval contains population mean μ.

Q) How many such $\bar{x}$ are there?

↳ 95% of all $\bar{x}$ are lying in the interval $\mu \pm \frac{2\sigma}{\sqrt{n}}$, hence all of them will have μ lying in the interval $\bar{x} \pm \frac{2\sigma}{\sqrt{n}}$ (by logic mentioned above).

↳ Hence 95% of sample ⇒ 0.95 probability ⇒ 95% confidence

CASE 6.1

Bankruptcy Attorney Fees In 2005, Congress passed the Bankruptcy Abuse Prevention and Consumer Protection Act. This act made significant changes to the administration of bankruptcy relief. The Government Accountability Office (GAO) was commissioned to study the effects of the bankruptcy reform law on consumers. Attorney fees in consumer bankruptcy cases were studied by the GAO and reported in their 2008 Bankruptcy Reform report.[2] In the period February to March 2007, there were 71,106 Chapter 7 consumer bankruptcy cases.*

To estimate legal fees for Chapter 7 consumer bankruptcy cases, the GAO conducted a nationwide random sample of Chapter 7 bankruptcy filings. Business bankruptcy cases and cases not involving attorneys were excluded. In the end, the GAO had $n = 292$ consumer

cases with attorney involvement. The mean attorney fee in the sample is $\bar{x} = \$1078$ and the standard deviation is $s = \$592$.

The sample mean of $1078 provides us with a single-value estimate of the population mean. However, by itself, this number gives us no information to assess just how close we might be to the population mean. A confidence interval tells us a much more complete story. Let us turn to its construction for this Case Study.

□ Example 6.6 : Confidence interval for bankruptcy Attorney Fees:

⇒ The study of example 6.1 is associated with a fairly large sample size.
↳ This allows us to use s as population σ.

∴ For 95% confidence, $z^* = 1.96$

∴ The margin of error,

$$z^* \frac{\sigma}{\sqrt{n}} = 1.96 \times \frac{592}{\sqrt{592}} = 67.9.$$

∴ A 95% confidence interval for μ is

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} = 1078 \pm 67.9$$
$$= (1010.1, 1145.9)$$

⇒ We are 95% confident that the mean attorney fees is between $1010.1 and $1145.9.

○ How confidence interval behave:

⇒ The margin of error illustrates several important properties that are shared by all confidence intervals in common use.

⇒ In particular, a higher confidence level increases z^* and so increases the margin of error for intervals based on same data (same std).
↳ We want high confidence & small margin of error.
↳ improving one degrades other.

⇒ If a confidence's margin of error is too large, there are only three ways to reduce it: $\left(z^* \frac{\sigma}{\sqrt{n}}\right)$
→ Use a lower level of confidence (smaller C, hence smaller z^*)
→ Reduce σ.
→ Increase sample size.

⇒ The common choices of confidence level are 99%, 95%, and 90%.

⇒ Sometimes we can reduce σ by carefully controlling the measurement process, or by restricting our attention to only a part of large population.

EXAMPLE 6.7 How Confidence Intervals Behave

CASE 6.1 Suppose there were only 100 attorney-involved Chapter 7 bankruptcy cases in the GAO study, and that $\bar{x}$ and σ are unchanged from Example 6.6. The margin of error increases from 67.9 to

$$z^* \frac{\sigma}{\sqrt{n}} = 1.96 \frac{592}{\sqrt{100}} = 116$$

A 95% confidence interval for μ is therefore

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} = 1078 \pm 116$$
$$= (962, 1194)$$

Figure 6.7 illustrates how the margin of error increases for the smaller sample.

Suppose next that we demand 99% confidence for the mean attorney fees in Example 6.6, rather than 95%. Table D tells us that for 99% confidence, $z^* = 2.576$. The margin of error

FIGURE 6.7 95% confidence intervals for $n = 100$ and $n = 292$, for Example 6.7.

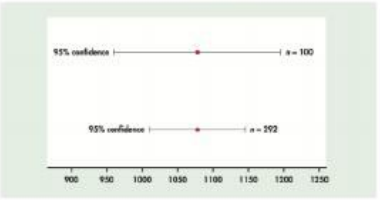
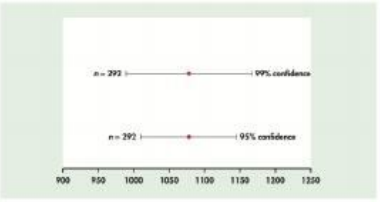


FIGURE 6.8 95% and 99% confidence intervals with $n = 292$ in both cases, for Example 6.7.



increases from 67.9 to

$$z^* \frac{\sigma}{\sqrt{n}} = 2.576 \frac{592}{\sqrt{292}}$$
$$= 89.2$$

The 99% confidence interval is therefore

$$\bar{x} \pm \text{margin of error} = 1078 \pm 89.2$$
$$= (988.8, 1167.2)$$

Requiring 99% confidence rather than 95% confidence has increased the margin of error from 67.9 to 89.2. Figure 6.8 compares the two intervals.

- Choosing the sample size :-

⇒ The margin of error of the confidence interval $\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$ for a Normal mean is

$$m = z^* \frac{\sigma}{\sqrt{n}}$$ (m: margin of error)

⇒ To obtain desired margin of error just set the expression equal to m, substitute the z^* for your confidence level & solve for sample size n.

⇒ The confidence interval for a population mean will have a specified margin of error m when the sample size is

$$n = \left(\frac{z^* \sigma}{m} \right)^2$$

→ It's the sample size, not the population size that determines the error, it make sense we're trying to estimate population.

EXAMPLE 6.8 How Many Bankruptcy Cases Should Be Sampled?

CASE 6.1 In Example 6.6, we found that the margin of error was 67.9 for estimating the mean attorney fees in Chapter 7 consumer bankruptcy cases with $n = 292$ and 95% confidence. If the GAO is willing to settle for a margin of error of $100 when it does a subsequent study, how many cases should the GAO sample?

For 95% confidence, Table D gives $z^* = 1.960$. Assume that the standard deviation in a future study will be approximately the same as the $\sigma = 592$ that the GAO obtained in its most recent study. For a margin of error of $100 we have

$$n = \left(\frac{z^* \sigma}{m} \right)^2 = \left(\frac{1.96 \times 592}{100} \right)^2 = 134.6$$

Because 134 observations will give a slightly wider interval than desired and 135 observations will give a slightly narrower interval, we recommend that the GAO have 135 randomly selected cases in the study. With this choice, the estimate of mean attorney fees will be within $100 of the true value with 95% confidence.

- Some cautions :-

⇒ We've already seen that small margins of error and high confidence can require large number of observations.

⇒ Because of the fact that "any formula for inference is correct only in specific circumstances".

⇒ Our formula $\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$ for estimating a normal mean comes with the following warnings :

1) The data must be SRS from the population.
   - → We are completely safe if we actually did a randomization & drew an SRS.
   - → Though we're not in great danger if the data can be plausibly treated as independent observations from a population.

2) The formula is not correct for probability sampling designs more complex than an SRS. (Not meant for multistage or stratified sample). Most Imp!

3) There is no correct method for inference from data haphazardly collected (badly produced data) with bias of unknown size.

4) Because $\bar{x}$ is not resistant, outliers can have large effect on the confidence interval.
   - → You should search for outliers, and try to correct them or justify their removal, before computing the interval.

5) If the sample size is small, and the population is not normal, the true confidence interval will be different from the value C used in computing the interval.
   - → The interval relies only on the sampling distribution of $\bar{x}$.
     - → Which even for small sizes is much closer to normal, when $n \geq 15$ the confidence interval is not greatly disturbed by non-Normal populations unless extreme outliers or strong skewness is present.

6) This only works if σ is known, if n is large then sample $\sigma \approx$ population sigma.

Finally, you should understand what statistical confidence does not say. We are 95% confident that the mean monthly HMO premium in Example 6.5 lies between $394.48 and $415.56. This says that these numbers were calculated by a method that gives correct results in 95% of all possible samples. It does *not* say that the probability is 95% that the true mean falls between $394.48 and $415.56. No randomness remains after we draw a particular sample and get from it a particular interval. The true mean either is or is not between $394.48 and $415.56. Probability in its interpretation as long-term relative frequency makes no sense in this situation. The probability calculations of standard statistical inference describe how often the *method* gives correct answers.

- Beyond the basics : The bootstrap :

⇒ Confidence intervals are based on sampling distribution.

⇒ The central limit theorem states that the sampling distribution is approximately $N(\mu, \sigma/\sqrt{n})$ for large sample from non-Normal populations.

Q) What if population is clearly non-Normal, and we only have a small sample?

↳ Then we don't know what the sampling distribution of $\bar{x}$ looks like.

A) The bootstrap is the procedure for approximating sampling distributions when theory cannot tell us their shape.

⇒ The basic idea is to act as if our sample were the population.

↳ We take many samples from it. Each of these is called a resample.

↳ For each resample, we calculate the mean $\bar{x}$.

⇒ We get different results from different resamples, because we sample with replacement.

The HMO premium data described in Example 6.5 are somewhat skewed toward high values. The mean \$405.02 is larger than the median, \$390.20. There are two firms with HMO premiums of \$1114.71 and \$1185 that could be considered outliers. Because we have no reason to suspect that these firms are qualitatively different from the other firms in the sample, we are reluctant to discard them. ~~Rather, we will reexamine our inference with a method that is not sensitive to the Normality assumption.~~

⇒ If we had 1000 SRSs from the population of firms, the distribution of the 1000 values of $\bar{x}$ would be close to the sampling distribution of $\bar{x}$.

⇒ We have only one sample (of size 453).

↳ The bootstrap says take 1000 samples (of size 453) from our one sample of size 453 from with replacement.

↳ For each resample, compute the mean $\bar{x}$. Treat the distribution of $\bar{x}$'s from the 1000 resamples as if it were the sampling distribution.

EXAMPLE 6.9 Use the Bootstrap to Check the Interval

In the discussion of Example 6.5 we found a 95% confidence interval for the mean HMO premium using the traditional statistical procedure based on the middle 95% of a Normal distribution. The interval was (394.48, 415.56). Let's check to see if we made a serious error. The middle 95% of the 1000 $\bar{x}$'s from 1000 resamples runs from 394.92 to 415.74. Try it again with 1000 new resamples: we get the interval (394.76, 415.52). The two bootstrap estimates are very close to each other and also very close to the standard interval that assumes Normality. The standard interval is reasonably accurate for these data.

If bootstrap method applied to this problem gave very different results, we need to reconsider the analysis.

↳ Obtaining similar results from different methods gives us confidence that the answers we've made are coming from the data, not from the particular methods that we used.

- Tests of significance :-

6.2

⇒ Confidence intervals is the most common type of formal statistical inference.

↳ They're appropriate when our goal is to estimate a population parameter.

⇒ The second common type of inference is directed at goal: to assess the evidence provided by the data in favor of some claim about the population.

• The reasoning of significance tests:

⇒ A significan test is a formal procedure for comparing observed data with a hypothesis whose truth we want to assess.

⇒ The hypothesis is a statement about the parameters in a population or a model.

⇒ The result of tests are measured in terms of probability that measures how well the data & the hypothesis agree.

CASE 6.2 **Fill the Bottles** Perhaps one of the ~~most common applications of hypothesis testing of the mean~~ is the ~~quality control problem~~ of assessing whether or not the underlying population mean is on "target." Consider the case of Bestea Bottlers. One of Bestea's most popular products is the 500 milliliters (ml) bottle of raspberry-flavored green tea. There is some variation from bottle to bottle because the filling machinery is not perfectly precise. Bestea has two concerns: whether there is a problem of underfilling (customers are then being shortchanged, which is a form of false advertising), or whether there is a problem of overfilling (unnecessary cost to the bottler).

⇒ In case 6.2, there is an intimate understanding of what is important to be discovered.

↳ In particular, is the population mean too high or too low relative to a desired level?

⇒ Let's proceed with the question of whether the bottling process is either underfilling or overfilling bottles.

EXAMPLE 6.10 Are the Bottles Being Filled as Advertised?

CASE 6.2

DATA FILE BOTTLEFILL

The filling process is not new to Bestea. Data on past production shows that the distribution of the contents is Normal with standard deviation $\sigma = 2$ ml. To assess the state of the bottling process, an inspector measures the contents of 10 randomly selected bottles. The results are

502.9 499.8 503.2 502.8 500.9 503.9 498.2 502.5 503.8 501.4

For this sample of 10 observations, the mean content ($\bar{x}$) is 501.94 ml. Is a sample mean of 501.94 ml convincing evidence that the mean fill of all bottles produced by the current process differs from the advertised 500 ml?

If we lack proper statistical thinking, this is a juncture to knee-jerk in one of two ways. One knee-jerk response is to say, "The mean of the bottles sampled is not 500 ml so the process is not filling the bottles at a mean level of 500 ml." Another knee-jerk response is to say, "The difference of 1.94 ml is small relative to the 500 ml baseline so there is nothing unusual going on here." ~~What these responses fail to do is consider the underlying variability of the population, which ultimately implies a failure to consider the sampling variability of the mean statistic.~~

To proceed, we step back and ~~suppose~~ that the truth about the population is $\mu = 500$ ml. ~~This is our hypothesis. We now ask, "*What is the probability of observing a simple mean at least as far from 500 ml as 1.94 ml?*" Taking into account sampling variability, the answer is 0.002.~~ (You will learn how to find this probability in Example 6.15.) ~~Because this probability is so small, we see that the sample mean $\bar{x} = 501.94$ is incompatible with a population mean of $\mu = 500$.~~ With this evidence, we are led to the conclusion that the bottling process does not have a mean of $\mu = 500$ ml. This suggests the need for corrective action.

↳ In other words "If the sample mean is 500 ml, what is the probability of observing this data".

↳ Same as saying what is the probability of observing our sample mean away from 500 ml atleast by 1.94 ml, if the Population mean is 500 ml

↳ Here the sample mean itself is 501.94, ie 1.94 ml away, hence all it is asking is

⇒ The fact that calculate probability is so small (0.002) leads us to believe that the mean fill amount is not in fact 500.

↳ Probability of observing the data if the mean fill was 500.

⇒ We calculated a probability assuming that the first probability is true, that is $\mu = 500$.

↳ That probability guide our final choice

↳ If the probability is very small, observing our data is highly unlikely, if the mean is equal to 500.

↳ Hence we conclude that mean is not in fact 500.

↳ There is always the chance that our conclusion is wrong.

EXAMPLE 6.11 Is It Right Now?

CASE 6.2

In Example 6.10, sample evidence suggested that the mean fill amount was not at the desired target of 500 ml. In particular, it appeared that the process was overfilling the bottles on average. In response, Bestea's production staff made adjustments to the process and collected a sample of 10 bottles from the "corrected" process. From this sample, we find $\bar{x} = 499.47$ ml. (We assume that the standard deviation is the same, $\sigma = 2$ ml.) In this case, the sample mean is less than 500 ml—to be exact, 0.53 ml less than 500 ml. Did the production staff overreact and adjust the mean level too low? We need to ask a similar question as in Example 6.10. In particular, what is the probability that the mean of a sample of size $n = 10$ from a Normal population with mean $\mu = 500$ and standard deviation $\sigma = 2$ is as far away or farther away from 500 as 0.53? The answer is 0.402. A sample result this far from 500 would happen just by chance in 40.2% of samples from a population having a true mean of 500. An outcome that could so easily happen just by chance is not convincing evidence that the population mean differs from 500. Bestea can feel confident that the corrections made to the filling process fixed the problem.

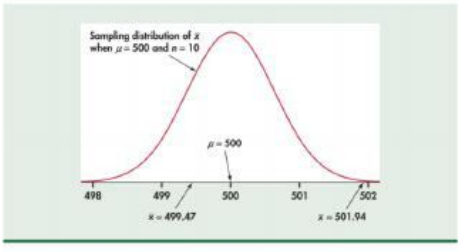


FIGURE 6.9 The mean fill amount for a sample of 10 bottles will have this sampling distribution if the mean for all bottles is $\mu = 500$. A sample mean $\bar{x} = 501.94$ is so far out on the curve that it would rarely happen just by chance.

⇒ The above figure 6.9 compares the two results graphically.

↳ It's the normal curve of sampling distribution $\bar{x}$, when $\mu = 500$.

↳ You can see that $\bar{x} = 501.94$ is an extreme outcome.

⇒ Here lies the core reasoning of statistical tests :

↳ A sample outcome that would be extreme if a hypothesis were true is an evidence that the hypothesis may not be true.

- Stating Hypothesis :-

⇒ The first test in significance test is to state a claim that will try to find evidence against.

- Null Hypothesis H_0 :-

⇒ The statement being tested in a test of significance is called null hypothesis.

→ The test of significance is designed to assess the strength of the evidence against the null hypothesis.

→ Usually the null hypothesis is a statement of "no effect" or "no difference".

⇒ A null hypothesis is a statement about a population, expressed in terms of some parameter or parameters.

ex:- for ex 6.10 & 6.11, the null hypothesis is

$$H_0 : \mu = 500.$$

⇒ It's convinient also to give name to the statement we hope or suspect to be true instead of H_0.

→ This is called the alternative hypothesis and is abbreviated as H_a.

→ In ex 6.10 & 6.11, the alternative hypothesis states that the mean fill amount is not 500. We write it as

$$H_a : \mu \neq 500.$$

Hypothesis always refer to some population or model, not to a particular outcome.

→ For this reason, we always state H_0 and H_a in terms of population parameters.

⇒ Because H_a expresses the effect that we hoped to find evidence for, we often begin with H_a then setup H_0 as the statement that the hoped for effect is not present.

⇒ Stating H_a is not straight forward.

→ The $H_a : \mu \neq 500$ in the bottle-filling examples is two-sided.

→ In both examples, we simply asked if mean fill amount is off target.

→ The process can be off target in that it fills too much, or too little.

⇒ Here is the setting where one sided alternative is appropriate.

EXAMPLE 6.12 Have We Reduced Processing Time?

Your company hopes to reduce the mean time μ required to process customer orders. At present, this mean is 3.8 days. You study the process and eliminate some unnecessary steps. Did you succeed in decreasing the average processing time? You hope to show that the mean is now less than 3.8 days, so the alternative hypothesis is one-sided, H_a: $\mu < 3.8$. The null hypothesis is as usual the "no-change" value, H_0: $\mu = 3.8$ days.

⇒ If you dont have any specific direction in mind in advance, use a two-sided alternative. Some statisticians argue that we should always work with the two-sided alternative.

⇒ The hypothesis (H_0 & H_a) accounts for all possible values of a parameter.
↳ Hence we could have wrote for example 6.12 as

H_0 : $\mu \geq 3.8$

H_a : $\mu < 3.8$

EXAMPLE 6.13 Bottle Fill Amount: The Hypotheses

CASE 6.2 For Example 6.10, the hypotheses are stated in terms of the mean fill amount for all bottles:

$$H_0: \mu = 500$$
$$H_a: \mu \neq 500$$

The estimate of μ is the sample mean $\bar{x}$. Because H_a is two-sided, values of $\bar{x}$ far from 500 on either the low or the high side count as evidence against the null hypothesis.

A test statistic compares the compatibility between the null hypothesis & the data.
↳ Many test statistic can be thought of as a distance between a sample estimate of a parameter & the value of parameter specified by the null hypothesis.

EXAMPLE 6.14 Bottle Fill Amount: The Test Statistic

CASE 6.2 For Example 6.10, the null hypothesis is H_0: $\mu = 500$, and a sample gave $\bar{x} = 501.94$. The test statistic for this problem is the standardized version of $\bar{x}$:

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

← Sample − Population
← std of sampling distribution
} z-value for sampling distribution because $\sim N(\mu, \frac{\sigma}{\sqrt{n}})$

This statistic is the distance between the sample mean and the hypothesized population mean in the standard scale of z-scores. In this example,

$$z = \frac{501.94 - 500}{2/\sqrt{10}} = 3.07$$

Even without a formal probability calculation, we know that standard score $z = 3.07$ lies far out on a Normal curve, which, in turn, suggests incompatibility of the observed sample result with the null hypothesis.

⇒ In applications where the distribution of the population is unknown, we can lean on central limit theorem.

- P-values:

⇒ In example 6.14, the standardized distance of the sample outcome $\bar{x}$ from the population mean $\mu = 500$ was $z = 3.07$. This suggest that the sample is not compatible with the null hypothesis.
↳ In fact, the supreme court of US has said that "two or three standard deviations" ($z = 2$ or 3) is its criterion for rejecting H_0.

⇒ Because not all test statistics produces z-scores, we translate the values of test statistics into a common language, the language of probability.

⇒ P-value: The probability, computed assuming H_0 is true, that the test statistic would take a value as extreme or more extreme than that actually observed is called P-value of the test.

- The smaller the P-value, the stronger evidence against H_0 provided by the data.
- If the test statistic has high probability to take as extreme value as observed (equal to) or more extreme (worst case / greater than) then null hypothesis is correct.
- Same as saying high probability that the data is from a population where null hypothesis is true.

⇒ To calculate the P-value, we must use the sampling distribution of the test statistic.

EXAMPLE 6.15 Bottle Fill Amount: The *P*-Value

CASE 6.2 In Example 6.10 the observations are an SRS of size $n = 10$ from a population of bottles with $\sigma = 2$. The observed average fill amount is $\bar{x} = 501.94$. In Example 6.14, we found that the test statistic for testing H_0: $\mu = 500$ versus H_a: $\mu \neq 500$ is

$$\therefore z = \frac{501.94 - 500}{2/\sqrt{10}} = 3.07$$

⇒ If the null hypothesis is true, we expect z to take values not too far from 0. Because the alternative is two-sided, values of z far from 0 in either direction count as evidence against H_0, and in favour of H_a

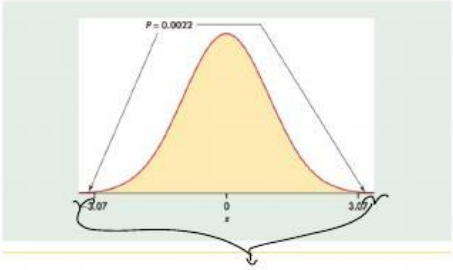
So the P-value (Probability of the observed data being generated by chance given H_0 is true) is

$$P(z \leq -3.07) + P(z \geq 3.07)$$ (example of two side / two tailed)

↳ other are (low tailed, high tailed) / one direction.

$$\therefore P = 0.0022$$

FIGURE 6.10 The P-value for Example 6.15. The two-sided *P*-value is the probability (when H_0 is true) that $\bar{x}$ takes a value at least as far from 0 as the actually observed value.



Only 0.002 % of the data are present after z & before −z, since we took two tailed & since our data lies there given that H_0 is true, H_0 is highly unlikely.

- Statistical Significance :

⇒ You must decide which test is appropriate & whether to use a one-sided or two-sided test.

⇒ We compare P-value with a level that says "this evidence is strong enough".
↳ The decisive level is called the significance level. (denoted by α)

⇒ If we choose $\alpha = 0.05$, we're requiring that the data give evidence against H_0 so strong that the data would happen (assuming H_0 to be true) no more than 5% of the time.

Statistically Significant : If the P-value is as small or smaller than α, we say that the data are statistically significant at level α.

⇒ You need not actually find the P-value to asses significance at a fixed level α. You need only compare the observed statistics z with critical value that marks of the area α in one or both tails of the standard Normal curve.

EXAMPLE 6.16 Significant?

CASE 6.2 In Example 6.14, the test statistic took the value $z = 3.07$. Is this significant at the $\alpha = 0.01$ level? Given the magnitude of the test statistic value, we are pretty certain about the answer to this question. Let us still walk through the thinking process. Values in the extreme 0.01 area of the standard Normal curve are significant at the 0.01 level. Because the alternative is two-sided, this area is divided equally between the two tails of the curve. So z is significant if it lies in the extreme area 0.005 at either end. Look at the z^* row at the bottom of Table D. The extreme 0.005 in the right tail starts at $z^* = 2.576$. That is, the z-values that are significant at the $\alpha = 0.01$ level are $z > 2.576$ and $z < -2.576$. Figure 6.11 shows why. The sample outcome $z = 3.07$ is thus statistically significant ($P < 0.01$).

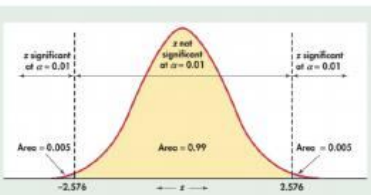


FIGURE 6.11 The P-value for Example 6.16. The two-sided P-value is the probability (when H_0 is true) that $\bar{x}$ takes a value at least as far from 0 as the actually observed value.

- Tests for a Population mean :

⇒ There are four steps to carry out a significance tests.

1) State your hypothesis
2) Calculate the test statistics.
3) Find the P-value.
4) State your conclusion in the context of your specified setting.

z Test for a Population Mean

To test the hypothesis H_0: $\mu = \mu_0$ based on an SRS of size n from a population with unknown mean μ and known standard deviation σ, compute the **one-sample z statistic**

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

In terms of a variable Z having the standard Normal distribution, the P-value for a test of H_0 against

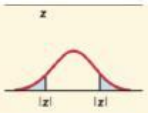
H_a: $\mu > \mu_0$ is $P(Z \geq z)$



H_a: $\mu < \mu_0$ is $P(Z \leq z)$



$H_a: \mu \neq \mu_0$ is $2P(Z \geq |z|)$



These P-values are exact if the population distribution is Normal and are approximately correct for large n in other cases.



EXAMPLE 6.17 Blood Pressures of Executives

The medical director of a large company is concerned about the effects of stress on the company's younger executives. According to the National Center for Health Statistics, the mean systolic blood pressure for males 35 to 44 years of age is 128 and the standard deviation in this population is 15. The medical director examines the records of 72 executives in this age group and finds that their mean systolic blood pressure is $\bar{x} = 129.93$. Is this evidence that the mean blood pressure for all the company's young male executives is higher than the national average? As usual in this chapter, we make the unrealistic assumption that the population standard deviation is known, in this case that executives have the same $\sigma = 15$ as the general population.

Step 1 Hypothesis: The hypothesis about the unknown mean μ of the executive population are

$H_0: \mu = 128$ (no effect of stress)

$H_a: \mu > 128$ (effect of stress)

Step 2 : Test statistics :

⇒ The z tests require that the 72 executives in the sample are an SRS from the population of the company's male executives.

∴ one sample ($\bar{x}$) z-statistic are

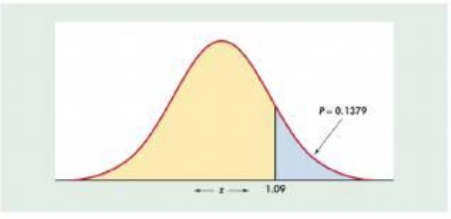
$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}} = \frac{129.93 - 128}{15/\sqrt{72}}$$

$$= 1.09.$$

Step 3: P-value

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FIGURE 6.12 The P-value for the one-sided test in Example 6.17.



⇒ P-value for our H_a is the probability that a standard Normal variable Z takes a value of 1.09 or greater.

$\therefore P = P(Z \geq 1.09) = 1 - 0.8621 = 0.13729$

Step 4 : conclusion

⇒ About 14 % of the time an SRS of size 72 from general male population ($\mu = 128$) would have mean ($\bar{x}$) = 129.93.

↳ Hence the data & H_0 are consistent, because 14 % is way higher than 5%, so sample data probably came from distribution where mean was μ (128).

Failing to find evidence against H_0 doesn't mean we have clear evidence that H_0 is true, it's just that data are consistent with H_0.

EXAMPLE 6.18 A Company-Wide Health Promotion

The company medical director institutes a health promotion campaign to encourage employees to exercise more and eat a healthier diet. One measure of the effectiveness of such a program is a drop in blood pressure. Choose a random sample of 50 employees, and compare their blood pressures from annual physical examinations given before the campaign and again a year later. The mean change in blood pressure for these $n = 50$ employees is $\bar{x} = -6$. We take the population standard deviation to be $\sigma = 20$.

Step 1: Hypothesis

⇒ We want to know if the health campaign reduced average blood pressure.

⇒ Take μ to be the mean change in blood pressure of all employees,

$H_0 : \mu = 0$

$H_a : \mu < 0$

Step 2: Test statistic : The one sample z-test is appropriate.

Assuming H_0 is true what is the z-score for the observed mean (the observed mean is a single sample in the sampling distribution)

$$\therefore z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{-6 - 0}{20/\sqrt{50}}$$

$$z = -2.12.$$

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Step 4: Conclusion

⇒ Since the p value is so small, the data is inconsistent with H_0, hence we found significant evidence for H_a

⇒ This is convincing evidence that the mean blood pressure in the population of all employees has decreased.

Two sided significance tests & confidence intervals :

⇒ A 95% confidence interval captures the true value of μ in 95% of all samples.

→ If we're 95% confident that the true μ lies in our interval, we're also confident that values of μ that falls outside the interval are incompatible with the data.

→ Same connections holds between 99% confidence intervals & significance at 1% level & so on.

A level α two-sided significance test rejects a hypothesis $H_0: \mu = \mu_0$ exactly when the value (α) μ_0 falls outside the $1-\alpha$ confidence interval for μ.

ex: 95% confidence ⇔ 5% significance level

EXAMPLE 6.19 Performance of IPO Firms

The decision to go public is clearly one of the most significant decisions to be made by a privately owned company. The first sale of stock to the public by a private company is referred to as an initial public offering (IPO). The natural question to ask is if the decision to go public is financially beneficial to the firm? In a study of a sample of 3964 IPO firms from a thirty-year period, it was found that the mean change in firm profitability, measured as return on equity (ROE),* was −4.29% over a three-year span starting from the initial offering.[5] The sample standard deviation s was 15.56. Clearly, the very large sample size allows us to safely use the sample standard deviation s as if it were the population standard deviation σ. In addition, the very large sample size guarantees that the sampling distribution of $\bar{x}$ is indistinguishable from the Normal distribution.

The researchers of this study asked whether the −4.29% mean change in ROE in the *sample* provides evidence that there is a nonzero mean change in ROE in the whole population of IPO firms?** This calls for a test of the hypotheses

$$H_0: \mu = 0$$
$$H_a: \mu \neq 0$$

We will carry out the test twice, first with the usual significance test and then with a 99% confidence interval.

1) The significance test:

⇒ The mean of the sample is −4.29, the $H_0: \mu = 0$

$$\therefore z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{-4.29 - 0}{15.56/\sqrt{3964}} = -17.36.$$

The p value is

$$P(z \leq -17.36) + P(z \geq 17.36)$$

⇒ because $H_0: \mu = 0$ & $H_a: \mu \neq 0$ ⇒ its two tailed / two sided

⇒ also the z-values that are significant at 1% are $z > 2.576$ & $z < -2.576$. Our $z = -17.36$ clearly satisfy these, so it's clearly significant ($P < 0.01$).

⇒ To compute a 99% confidence interval for the mean change in ROE.

→ ∴ The critical value z^* for 99% confidence is $z^* = 2.576$

∴ the interval is $\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}}$

$$= -4.29 \pm 2.576 \frac{15.56}{\sqrt{3964}}$$

U3964

$= -4.29 \pm 0.6366$

$= (-4.927, -3.653)$

- The $H_0 : \mu = 0$ falls outside this confidence interval, hence we're therefore 99% confident that μ is not equal to 0, so we can reject $H_0 : \mu = 0$ at 1% significance level.

- On other hand we cannot reject the values of μ that fall in the range -4.927 to -3.653.
  - For example, $H_0 : \mu = -3.9$ cannot be rejected.

- P-values vs fixed α :

- For example 6.19, we concluded that the test statistic $z = -17.36$ is significant at the 1% level.
  - The observed z is far beyond the critical value for $\alpha = 0.01$ and the evidence against H_0 is far stronger than 1% significance suggests.
    - The P value was $P = 1.657 \times 10^{-67}$, so the evidence was astronomically strong against the hypothesis $\mu = 0$.

- In general the p-value is the smallest level α at which the data are significant.
  - Knowing p-values allows us to access significance at any level.

EXAMPLE 6.20 How Significant?

In Example 6.18, we tested the hypotheses

$$H_0: \mu = 0$$
$$H_a: \mu < 0$$

to evaluate the effectiveness of a health promotion program on blood pressure. The test had the P-value $P = 0.0170$. This result is not significant at the $\alpha = 0.01$ level, because $0.0170 \geq 0.01$. However, it is significant at the $\alpha = 0.05$ level, because the P-value is less than 0.05. See Figure 6.15.

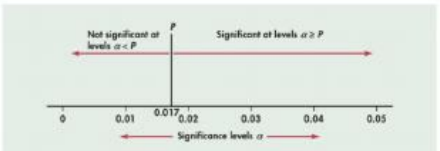


FIGURE 6.15 An outcome with P-value P is significant at all levels α above P and is not significant at smaller levels of α.

- Using Significance tests :

Significance tests are widely used in reporting the results of research in many fields of applied science and in industry. New pharmaceutical products require significant evidence of effectiveness and safety. Courts inquire about statistical significance in hearing class action discrimination cases. Marketers want to know whether a new ad campaign significantly outperforms the old one, and medical researchers want to know whether a new therapy performs significantly better. In all these uses, statistical significance is valued because it points to an effect that is unlikely to occur simply by chance.

- How small a P is convincing ?

- The purpose of the test of significance is to assess degree of evidence

provided by the sample against null hypothesis.

⇒ But how small of a P-value is convincing evidence against null hypothesis?
This depends on two circumstances.

→ How plausible is H_0? If H_0 represents an assumption that the people you must convince have believed for years, strong evidence (small P) will be needed to persuade them.

→ What are the consequences of rejecting H_0?

→ If rejecting H_0 in favor of H_a means making an expensive changeover from one type of product packaging to another, you'll need strong evidence that the new packaging will boost sales.

⇒ Standard levels of significance are 10%, 5%, 1%.

⇒ There's no practical distinction between P-values 0.049 & 0.051. It makes no sense to treat $P \leq 0.05$ as a universal rule for what is more significant.

Statistical significance and practical significance

When a null hypothesis ("no effect" or "no difference") can be rejected at the usual levels, $\alpha = 0.05$ or $\alpha = 0.01$, there is good evidence that an effect is present. But that effect may be extremely small. When large samples are available, even tiny deviations from the null hypothesis will be significant.

EXAMPLE 6.21 It's Significant. So What?

We are testing the hypothesis of no correlation between two variables. With 1000 observations, an observed correlation of only $r = 0.08$ is significant evidence at the $\alpha = 0.01$ level that the correlation in the population is not zero but positive. The low significance level does not mean there is a strong association, only that there is strong evidence of some association. The true population correlation is probably quite close to the observed sample value, $r = 0.08$. We might well conclude, however, that for practical purposes we can ignore the association between these variables, even though we are confident (at the 1% level) that the correlation is positive.

⇒ Statistical significance is not the same as practical significance.

⇒ A few outlying observations can produce highly significant results if you blindly apply common tests of significance.

⇒ It's usually wise to give a confidence interval for the parameter in which you're interested.

→ A confidence interval actually estimates the size of the effect rather than simply asking if it's too large to reasonably occur by chance alone.

- Statistical inference is not valid for all sets of data.

⇒ Tests of significance & confidence intervals are based on the laws of probability.

→ Randomization in sampling or experimentation ensures that these laws apply. But we must often analyze data that do not arise from randomized samples or experiments.

→ To apply statistical inference to such data, we must have confidence in a probability model for the data.

- Basics of testing for significance

⇒ **EXAMPLE 6.22** Cell Phones and Brain Cancer

Might the radiation from cell phones be harmful to users? Many studies have found little or no connection between using cell phones and various illnesses. Here is part of a news account of one study:

> *A hospital study that compared brain cancer patients and a similar group without brain cancer found no statistically significant association between cell phone use and a group of brain cancers known as gliomas. But when 20 types of glioma were considered separately an association was found between phone use and one rare form. Puzzlingly, however, this risk appeared to decrease rather than increase with greater mobile phone use.*[12]

Think for a moment: Suppose that the 20 null hypotheses for these 20 significance tests are all true. Then each test has a 5% chance of being significant at the 5% level. That's what $\alpha = 0.05$ means—results this extreme occur only 5% of the time just by chance when the null hypothesis is true. Because 5% is 1/20, we expect about 1 of 20 tests to give a significant result just by chance. ~~Running 1 test and reaching the $\alpha = 0.05$ level is reasonably good evidence that you have found something; running 20 tests and reaching that level only once is not.~~

⇒ The proper use of significant tests involves choosing one hypothesis before searching your data & then testing that one hypothesis.

- Power & inference as a decision :-

⇒ Although we prefer to use the P-values rather than the reject-or-not view of the fixed α significance test, the latter view is important for planning statistical studies.

- The power of statistical studies :

⇒ High confidence is of little value if the interval is so wide that few values of the parameters are excluded.

⇒ Similarly, a test with smaller α is of little value if it almost never rejects H_0 even when the true parameter value is far from the hypothesized value.

↳ We must be concerned with the ability to detect that H_0 is false, just as we're concerned with the margin of error of a confidence interval.

↳ This ability is measured by the probability that the test will reject H_0 when an alternative is true.

↳ The higher this probability is, the more sensitive the test is.

⇒ Power : The probability that a fixed level α significance test will reject H_0 when a particular alternate value of the parameter is true is called the power of the test.

EXAMPLE 6.23 Exercise and Bones

Can a 6-month exercise program increase the total body bone mineral content (TBBMC) of young women? A team of researchers is planning a study to examine this question. Based on the results of a previous study, they are willing to assume that $\sigma = 2$ for the percent change in TBBMC over the 6-month period. A change in TBBMC of 1% would be considered important, and the researchers would like to have a reasonable chance of detecting a change this large or larger. Are 25 subjects a large enough sample for this project?

⇒ We'll answer this question by calculating the power of the significant test.

⇒ The calculation consists of three steps.

1. State H_0, H_a, and the significance level α.
2. Find the value of $\bar{x}$ that will lead us to reject H_0.
3. Calculate the probability of observing these values of $\bar{x}$ when the alternative is true.

Step 1: The null hypothesis is exercise have no effect on TBBMC.

↳ i.e mean percent change is zero.

⇒ Formally, we have

$H_0: \mu = 0$

$H_a: \mu > 0$

⇒ Even though our alternative hypothesis considers μ value greater than 0, we're specifically interested in detecting a μ value of atleast 1.

↳ Well perform our power calculations with the conservative choice of $\mu = 1$.

↳ If in reality μ departs even further from H_0 value of 0 than 1, then the actual power will be even greater than what we would calculate for $\mu = 1$.

⇒ A 5% significance test will be used.

Step 2 The single sample z statistics is appropriate

Assuming H_0 is true,

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

⇒ The z-test rejects H_0 at $\alpha = 0.05$ level when

$$z \geq 1.645$$

$$\frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{\bar{x} - 0}{2/\sqrt{25}} \geq 1.645$$

⇒ Rewrite this in terms of $\bar{x}$.

rejects H_0 when $\bar{x} \geq 0.658$.

⇒ Because the significant level is $\alpha = 0.05$, this event ($\bar{x} \geq 0.658$ & H_0 being true) has probability of 0.05 of occuring when the population mean $\mu = 0$.

Step 3:

⇒ The power of the alternative $\mu = 1$ is the probability that H_0 will be rejected when in fact $\mu = 1$.

⇒ We calculate this probability by standardizing $\bar{x}$, using the value $\mu = 1$, the population standard deviation $\sigma = 2$ & the sample size $n = 25$

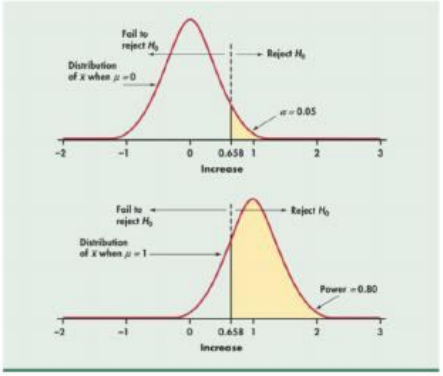
∴ The power is

$$P(\bar{x} \geq 0.658 \text{ when } \mu = 1) = P\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \geq \frac{0.658 - 1}{2/\sqrt{25}}\right)$$

} Basically assuming H_a is true

$$= P(Z \geq -0.855) = 0.80$$

FIGURE 6.16 The sampling distributions of $\bar{x}$ when $\mu = 0$ and when $\mu = 1$, with α and the power. Power is the probability that the test rejects H_0 when the alternative is true.



⇒ Along with 95% confidence intervals, 5% significance test, 80% power is becoming standard.

⇒ This significant test rejects H_0 80% of the time if the true effect of exercise is a 1% increase in TBBMC.

↳ If the true effect of exercise is greater than 1%, the test will have greater power & reject H_0 with higher probability.

- Increasing the Power:

1 Increase α: A 5% test of significance will have a greater chance of rejecting the null hypothesis than a 1% test, because the strength of evidence required for rejection is less.

2 Consider a particular alternative that is further from μ_0. Values of μ that are in H_a but lies close to the hypothesized value μ_0 are harder to detect (lower power) than the values of μ that are far from μ_0.

3 Increase the size of sample. More data will provide more information about $\bar{x}$ so we have a better chance of distinguishing values of μ.

4 Decrease σ. This will the same effect as increasing the sample size: more information about μ.

⇒ Power calculations are important in planning studies. Using significant test with low power makes it unlikely that you will find a significant effect even if the truth is far from the null hypothesis.

EXAMPLE 6.24 Are Stock Markets Efficient?

The "efficient market hypothesis" for stock prices says that future stock prices show only random variation. No information available now will help us predict stock prices in the future, because the efficient working of the market has already incorporated all available information in the present

price. Many studies tested the claim that one or another kind of information is helpful. In these studies, the efficient market hypothesis is H_0, and the claim that prediction is possible is H_a. Almost all studies failed to find good evidence against H_0. As a result, the efficient market theory became quite popular.

An examination of the significance tests employed found that the power was generally low. Failure to reject H_0 when using tests of low power is not evidence that H_0 is true. As one expert said, "The widespread impression that there is strong evidence for market efficiency may be due just to a lack of appreciation of the low power of many statistical tests."[13] More careful studies later showed that the size of a company and measures of value such as the ratio of stock price to earnings do help predict future stock price movements.

⇒ Here is another example of a power calculation, this time for a two-sided z test.

EXAMPLE 6.25 Find the Power for a Two-Sided *z* Test

CASE 6.2 Case 6.2 considered the following competing hypotheses:

$$H_0: \mu = 500$$
$$H_a: \mu \neq 500$$

In Example 6.10, we learned that $\sigma = 2$ ml for the filling process. Suppose that the bottler Bestea wishes to conduct tests of the filling process mean at a 1% level of significance. Assume, as in Example 6.10, that 10 bottles are randomly chosen for inspection. What is the power of this test against the specific alternative $\mu = 498.5$?

⇒ for 1% significance level the z-values are $z < -2.576$ & $z > 2.576$ which would be classed as significant at 1% level.

↳ ∴ The test reject H_0 when $|z| \geq 2.576$.

∴ The test statistic

$$z = \frac{\bar{x} - 500}{2/\sqrt{10}}$$

⇒ The different z's are disjoint events, so the power is sum of their probability computed assuming H_a is true, i.e $\mu = 498.5$ is true.

when $z \geq 2.576 \Rightarrow \bar{x} \geq 501.629$

$$\therefore P(\bar{x} \geq 501.629) = P\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \geq \frac{501.629 - 498.5}{2/\sqrt{10}}\right).$$

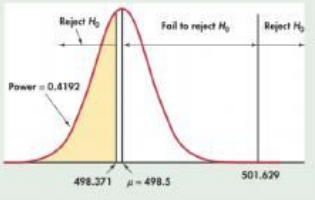
$$= P(z \geq 4.95) = 0$$

$$P(\bar{x} \leq 498.371) = P\left(\frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq \frac{498.371 - 498.5}{2/\sqrt{10}}\right)$$

$$= P(z \leq -0.204) = 0.4192.$$

⇒ Because the power is so low, we are not strongly confident that the test will reject H_0 when alternative is true.

FIGURE 6.17 Power for Example 6.25.



EXAMPLE 6.26 Choosing Sample Size for a Desired Power?

CASE 6.2 Suppose the bottler Bestea desires a power of 0.9 in the detection of the specific alternative of $\mu = 498.5$. From Example 6.25, we found that a sample size of 10 offers a power of only 0.4192. Manually, we can repeat the calculations found in Example 6.25 for different values of n larger

than 10 until we find the smallest sample size giving at least a power of 0.9. Fortunately, most statistical software saves us from such a tedium. Below is Minitab output with inputs of 0.9 for power, 1% for significance level, 2 for σ, and -1.5 ($= 498.5 - 500$) for the departure amount from the null hypothesis:

```
Power and Sample Size

1-Sample Z Test

Testing mean = null (versus not = null)
Calculating power for mean = null + difference
Alpha = 0.01  Assumed standard deviation = 2

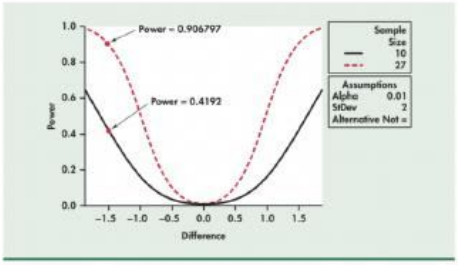
           Sample  Target
Difference   Size   Power  Actual Power
      -1.5     27     0.9      0.906797
```

From the output, we learn that a sample size of at least 27 is needed to have a power of at least 0.9. If we used a sample size of 26, the actual power would be less than the target power of 0.9.

⇒ In addition to the output shown in the above figure, statistical software will often show graphically the power of the test for several possible departures from null hypothesis.

↳ Such a graph is called as a power curve.

FIGURE 6.18 Power curves for two departures from the null hypothesis mean for a population with $\sigma = 2$ for a testing procedure using a 1% significance level and $n = 10$ and 27. The power values found in Examples 6.25 and 6.26 are labeled on the curves.



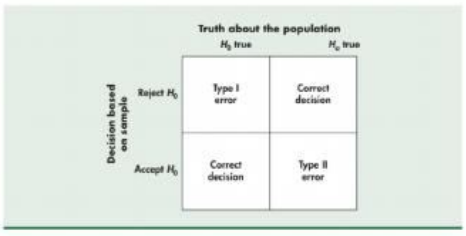
⇒ The curves are intuitive because we find that the power of a test for a given sample size becomes stronger the larger the departure (down or high) is from null hypothesis.

° Two types of error :

Type 1 & Type 2 error :

⇒ If we reject H_0 (accept H_a) when in fact the H_0 was true, this is called as type-1 error.
If we accept H_0 (reject H_a) when in fact the H_a was true, this is called a type-2 error.

FIGURE 6.19 The two types of error in testing hypotheses.



⇒ In case of machine learning, assume you have two types of classes,

class +ve (class 0): alternative hypothesis, the patient has the disease, the email is a spam.

↳ This is equivalent to alternative hypothesis

class -ve (class 1): not email, not disease ⇒ null hypothesis

Decision \ truth	+ve	−ve
+ve	✓	FP (type-1 error)
−ve	FN (type-2 error)	✓

actually −ve but pred +ve ⇒ false positive

↳ Null hypothesis was correct but you chose alternative hypothesis

- Error Probabilities:

⇒ We can access any rule for making decision in terms of the probabilities of the two types of error.

↳ We cannot guarantee that a bad decision won't be made based on them.

↳ But by random sample & the laws of probability, we can say what the probabilities of both kinds of errors are.

⇒ Significant tests with a fixed significance gives a rule for making decisions because the test either rejects H_0 or fails to reject it.

↳ We can then describe the performance of a test by probabilities of type-1 and type-2 errors.

EXAMPLE 6.27 Diameters of Bearings

The diameter of a particular precision ball bearing has a target value of 20 millimeters (mm) with tolerance limits of ±0.001 mm around the target. Suppose that the bearing diameters vary Normally with standard deviation of sixty-five hundred-thousandths of a millimeter, that is, $\sigma = 0.00065$ mm. When a shipment of the bearings arrives, the consumer takes an SRS of 5 bearings from the

[redacted]

shipment and measures their diameters. The consumer rejects the bearings if the sample mean diameter is significantly different from 20 mm at the 5% significance level.

This is a test of the hypotheses

$$H_0: \mu = 20$$
$$H_a: \mu \neq 20$$

To carry out the test, the consumer computes the z statistic:

$$z = \frac{\bar{x} - 20}{0.00065/\sqrt{5}}$$

and rejects H_0 if

$$z < -1.96 \text{ or } z > 1.96$$

⇒ A type-1 error is to reject H_0 when in fact $\mu = 20$

Q) What about type 2 error?

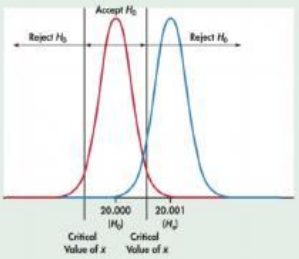
⇒ Because there are many values of μ in H_a, will concentrate on one value.

↳ Based on tolerance limits, the producer agrees that if there's evidence that the mean of ball bearings in the lot of 0.001 mm away from the desired mean of 20 mm, then the whole shipment should be rejected.

↳ So, a particular type-2 error is to accept H_0 when in fact $\mu = 20 + 0.001 = 20.001$

Figure 6.21 shows how the two probabilities of error are obtained from the two sampling distributions of $\bar{x}$, for $\mu = 20$ and for $\mu = 20.001$. When $\mu = 20$, H_0 is true and to reject H_0 is a Type I error. When $\mu = 20.001$, accepting H_0 is a Type II error. We will now calculate these error probabilities.

FIGURE 6.21 The two error probabilities for Example 6.27. The probability of a Type I error is the probability of rejecting H_0: $\mu = 20$ when in fact $\mu = 20$. The probability of a Type II error is the probability of accepting H_0 when in fact $\mu = 20.001$.



- Significance and type-1 error :-

⇒ The significance level α of any fixed level test is the probability of a type 1-error.

↳ That is α is the probability that the test will be reject the null hypothesis H_0 when H_0 is in fact true.

↳ Basically p-value is the probability that a random chance generated the data given H_0 is true.

↳ We reject the H_0 with probability equal to p value, i.e whenever $p \leq \alpha$ we reject the hypothesis, so if H_0 was actually true given the p-value then it is an error. Hence we say α to be type-1 error.

⇒ The probability of a type-2 error for a particular alternative $\mu = 20.001$ is the probability that the test will fail to value $\mu = 20.001$.

↳ The power of the test for the alternative $\mu = 20.001$ is just the probability that the test will reject the hypothesis given the alternative is true (i.e H_a is true).

↳ Hence 1 - power is the probability that the test doesn't reject the null hypothesis given H_a is true, i.e type-2 error.

- Power & type 2 error.

⇒ The power of a fixed level test for a particular alternative is 1 - (the probability of the type 2 error) for that alternative.

- The common practice of hypothesis testing :-

⇒ The common practice of testing hypothesis mixes the reasoning of significance tests and decision rules as follows :-

1) State H_0 & H_a just as in a test of significance

2) Think of the problem as decision problem, so that the probabilities of type-1 & type-2 errors are relevant.

3) Because of step 1, type 1 errors are more serious. So choose an α & consider only tests with type-1 error no greater than α.

4) Among these tests, select one that make probability of type 2 error as small as possible (that is, power as large as possible). If this probability of type 2 error is large, you will have to take a larger sample to reduce the chance of an error.